

CSEN 1144: Generative and Agentic AI, Spring Term 2026
Milestone 3
Due May 22, 2026

1. Milestone 3 Description

In Milestone 1, you have fine-tuned a model that generates 17 hypotheses' labels with their accompanying evidence spans for each supplied contract from ContractNLI dataset. Next, in Milestone 2, you have prepared for Milestone 3 by designing and partially implementing a multi-agent system by implementing one agent, a RAG pipeline and a GraphRAG pipeline.

In this milestone, you are first required to implement your designs after careful consideration of your feedback during evaluation sessions. The main objective is to compare and contrast the *multi-agent architecture* to *fine-tuned models* in terms of results, time-to-develop, and effort exerted in both using the same test-split from Milestone 1.

2. Requirements

To clearly address the difference in how *LLMs* and *agents* are used, you are required to implement an *agentic* system that could include a few modules of LLM calls and code, but must have multiple Agents that have feedback, memory, and tool calls. Your multi-agent system should satisfy the following.

- a) **17 Hypotheses Analysis mode.** Given a contract, the system analyzes it against the 17 hypotheses in ContractNLI dataset, outputting each hypothesis, label, & evidence in JSON format.
- b) **Conversation Mode (legal assistant)** A user of this system is able to upload a contract and ask generic questions about it and receive insightful, correct, and cited answers with external sources if any were used.
- c) **Auditability** across all agents and modules through the runtrace schema.
 - One runtrace per contract.
 - One runtrace per conversation session.
- d) **Application of playbook from Milestone 1.** You are required to integrate the playbook from Milestone 1 (no edits) in the process of the multi-agent system.
- e) **Usage of the system** is done through the CLI with flags and ability to converse (not just send one message.)
- f) Do not use the fine-tuned model in your implementation, you can use any one or more of the models from the same family of the fine-tuned model (e.g. you can use QWEN2.5:72B if you used a QWEN2.5 model for fine-tuning)
- g) You are required to have Vector RAG **and** GraphRAG in your implementation; however:

- You are at liberty to employ them however they fit your design.
 - You can change MS2 implementation to fit that new design.
- h) The runtrace MUST include the tool calls (Name, args, output and count) per agent.

3. System Evaluation

After finishing the above requirements, run an analysis on the same testing split you used for Milestone 1 and collect the same exact metrics.

- a) A predicted label for each hypothesis,
- b) Supporting evidence spans extracted from the contract,
- c) Deterministic policy outputs derived from the playbook rules,
- d) Traceable execution logs recorded in the RunTrace.
- e) Aggregate and collect the same metrics as Milestone 1.

The emphasis of this milestone is not only system accuracy, but also to conduct an imperial comparison between fine-tuning a model and constructing a multi-agent system for this use case.

4. Milestone Evaluation

Make sure that you understand your system, and how it came to exist. You will be asked both technical and non-technical questions about your implementation. You have to understand the implemented code inside-and-out and be able to explain it and defend and rationalize your design choices.

5. Deliverables

Required Deliverables

You must submit the following.

- a) Codebase (a new branch in the same github repo.)
- b) One final evaluation CSV (it includes both MS1 and MS3 metrics.)
- c) Zip file including all runtraces from the evaluation.
- d) A contribution markdown file describing each group member's contributions to the project.